King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam I Term 221

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The curve $y = \int_1^x \frac{t^2+1}{\sqrt{t}}\,dt$ is concave up on the interval:

(a) $(1, \infty)$

(b) $(-\infty, \sqrt{3})$

(c) $(0, 1)$

(d) $(0, \sqrt{3})$

(e) $\left(\frac{1}{2}, \infty\right)$

2. If $x\sin(\pi x) = \int_0^{x^2} f(t)\,dt$, where f is a continuous function, then $f(4) =$

(a) $\frac{2}{3}\pi$

(b) $\frac{3}{2}\pi$

(c) $\frac{\pi}{2}$

(d) 2π

(e) 2

3. The volume of the solid obtained by rotating the region bounded by the curves $x = \sqrt{y}$, $x = 0$ and $y = 1$ about $y = 1$ is

(a) $\frac{3}{5}\pi$

(b) $\frac{7}{13}\pi$

(c) $\frac{8\pi}{15}$

(d) $\frac{13\pi}{15}$

(e) $\frac{11\pi}{15}$

4. $\int_0^5 \left(x - \sqrt{25 - x^2}\right) dx =$

(a) $\dfrac{40 - 25\pi}{4}$

(b) $\dfrac{25 - 25\pi}{2}$

(c) $\dfrac{50 + \pi}{4}$

(d) $\dfrac{50 - 25\pi}{4}$

(e) $\dfrac{25 - 50\pi}{2}$

5. $\int_1^5 \frac{x\,dx}{\sqrt{2x-1}} =$

(a) $\dfrac{11}{3}$

(b) $\dfrac{13}{3}$

(c) $\dfrac{16}{3}$

(d) $\dfrac{8}{3}$

(e) $\dfrac{14}{3}$

6. $\int_0^{3\pi/2} |\sin x|\, dx =$

(a) -1

(b) 2

(c) 3

(d) 1

(e) -2

7. If $\int_0^4 e^{(x-2)^4}\, dx = k$, then the value of $\int_0^4 x\, e^{(x-2)^4}\, dx =$

(a) k^4

(b) $4k^2$

(c) k^2

(d) $2k$

(e) k

8. Find the area of the region enclosed by the curves $y = x^2$ and $y = 2 - |x|$.

(a) 7/3

(b) 7/6

(c) 5/3

(d) 13/3

(e) 5/6

9. The volume of the solid obtained by rotating the region bounded by the curves $y = 6 - x^2$ and $y = 2$ about $y = 2$ is

(a) $\pi \int_{-2}^{2} \left[(6 - x^2)^2 - 4\right] dx$

(b) $2\pi \int_{0}^{2} \left[(4 - x^2)^2 - 4\right] dx$

(c) $\pi \int_{-2}^{2} (x^2 - 4)\, dx$

(d) $2\pi \int_{0}^{2} (x^2 + 4)^2\, dx$

(e) $2\pi \int_{0}^{2} (4 - x^2)^2\, dx$

10. $\lim_{n\to\infty} \frac{3}{n} \sum_{i=1}^{n} \left[\left(\frac{3i}{n}\right)^3 - 6\left(\frac{3i}{n}\right)\right] =$

(a) $\dfrac{29}{4}$

(b) $\dfrac{27}{4}$

(c) $-\dfrac{27}{4}$

(d) $-\dfrac{26}{4}$

(e) $-\dfrac{29}{4}$

11. Find the volume of the solid obtained by rotating the region bounded by the curves $y = x^2$ and $x = y^2$ about the line $y = 1$.

(a) $\dfrac{\pi}{3}$

(b) $\dfrac{13}{30}\pi$

(c) $\dfrac{\pi}{6}$

(d) $\dfrac{11}{30}\pi$

(e) $\dfrac{\pi}{2}$

12. $\int_1^2 \frac{(x-1)^3}{x^2}\,dx =$

(a) $2\ln 3 - \sqrt{2}$

(b) $-2 + 3\ln 2$

(c) $\dfrac{3}{2} + \ln 2$

(d) $1 - 2\ln 2$

(e) $\ln 9$

13. Find the number b such that the line $y = b$ divides the region bounded by the curves $y = x^2$ and $y = 4$ into two equal halves.

(a) 2

(b) $2\sqrt[3]{4}$

(c) 8

(d) $2\sqrt[3]{2}$

(e) 4

14. $\int \frac{\cos\left(\frac{\pi}{x}\right)}{x^2}\,dx =$

(a) $-\dfrac{1}{x}\sin(\pi x) + c$

(b) $-\dfrac{x}{\pi}\sin\left(\dfrac{\pi}{x}\right) + c$

(c) $-\dfrac{1}{\pi}\sin(\pi x) + c$

(d) $\sin\left(\dfrac{\pi}{x}\right) + c$

(e) $-\dfrac{1}{\pi}\sin\left(\dfrac{\pi}{x}\right) + c$

15. The volume of the solid obtained by rotating the region bounded by the curves $y = \ln x$, $y = -1$ and $y = 1$, about $x = -1$ is given by

(a) $2\pi \int_0^1 \left[(1 + \ln x)^2 - 1\right] dx$

(b) $2\pi \int_0^1 \left(e^{2y} + 2e^y\right) dy$

(c) $\pi \int_{-1}^1 \left[(1 + \ln x)^2 - e^x\right] dx$

(d) $\pi \int_{-1}^1 \left(e^{2y} + 2e^y\right) dy$

(e) $\pi \int_{-1}^1 \left[(1 + e^y)^2 - e^{2y}\right] dy$

16. $\dfrac{d}{dx}\left[\int_{1-3x}^{1} \frac{u^3}{1+u^2}\,du\right] =$

(a) $\dfrac{3(1-3x)^3}{1+(1-3x)^2}$

(b) $\dfrac{(1-3x)^3}{1+(1-3x)^2}$

(c) $\dfrac{3x^3}{1+x^2}$

(d) $\dfrac{-3(1-3x)^3}{1+(1-3x)^2}$

(e) $\dfrac{1}{2}$

17. $\int_1^3 \sqrt{4+x^2}\,dx =$

(a) $\lim_{n\to\infty} \sum_{i=1}^{n} \left(\sqrt{4+\frac{4i^2}{n^2}}\right) \frac{2}{n}$

(b) $\lim_{n\to\infty} \sum_{i=1}^{n} \left(\sqrt{4+\left(1+\frac{i}{n}\right)^2}\right) \frac{1}{n}$

(c) $\lim_{n\to\infty} \sum_{i=1}^{n} \left(\sqrt{4+\left(1-\frac{2i}{n}\right)^2}\right) \frac{2}{n}$

(d) $\lim_{n\to\infty} \sum_{i=1}^{n} \left(\sqrt{4+\left(1+\frac{2i}{n}\right)^2}\right) \frac{2}{n}$

(e) $\lim_{n\to\infty} \sum_{i=1}^{n} \left(\sqrt{4+\left(1+\frac{2i}{n}\right)}\right) \frac{2}{n}$

18. If U and L are upper and lower sums for $f(x)=x^2$, $0\le x\le 2$, with $n=4$ then $\dfrac{U+L}{2} =$

(a) $\dfrac{11}{4}$

(b) 3

(c) $\dfrac{7}{4}$

(d) $\dfrac{13}{4}$

(e) $\dfrac{15}{4}$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$y'' = \frac{3x^2-1}{2x^{3/2}} > 0$ for $x > \frac{1}{\sqrt{3}}$, which contains $(1, \infty)$.
2	**(c)**	Differentiate, set $x = 2$: $2\pi = 4f(4)$, so $f(4) = \pi/2$.
3	**(c)**	Disks of radius $1 - x^2$: $\pi \int_0^1 (1 - x^2)^2 dx = \frac{8\pi}{15}$.
4	**(d)**	$\frac{25}{2} - \frac{25\pi}{4} = \frac{50-25\pi}{4}$ (triangle minus quarter disk).
5	**(c)**	$u = \sqrt{2x-1}$: $\frac{1}{2}\int_1^3 (u^2+1)du = \frac{16}{3}$.
6	**(c)**	Split at π: $\int_0^\pi \sin x\, dx + \int_\pi^{3\pi/2} (-\sin x)\, dx = 2 + 1 = 3$.
7	**(d)**	$u = x - 2$: odd part vanishes, $2\int_{-2}^2 e^{u^4} du = 2k$.
8	**(a)**	Intersections $x = \pm 1$: $2\int_0^1 (2 - x - x^2)dx = \frac{7}{3}$.
9	**(e)**	Disks of radius $4 - x^2$ on $[-2, 2]$; by symmetry $V = 2\pi \int_0^2 (4 - x^2)^2 dx$.
10	**(c)**	Riemann sum for $\int_0^3 (x^3 - 6x)dx = \frac{81}{4} - 27 = -\frac{27}{4}$.
11	**(d)**	Washers about $y = 1$: $\pi \int_0^1 [(1 - x^2)^2 - (1 - \sqrt{x})^2]dx = \frac{11\pi}{30}$.
12	**(b)**	Expand: $x - 3 + \frac{3}{x} - \frac{1}{x^2}$; antiderivative evaluated gives $-2 + 3\ln 2$.
13	**(d)**	$\frac{2}{3}b^{3/2} = \frac{2}{3}(8 - b^{3/2}) \Rightarrow b^{3/2} = 4$, $b = 2\sqrt[3]{2}$.
14	**(e)**	$u = \pi/x$, $\frac{dx}{x^2} = -\frac{du}{\pi}$: result $-\frac{1}{\pi}\sin(\pi/x) + c$.
15	**(d)**	Washers about $x = -1$: $R = 1 + e^y$, $r = 1$, $V = \pi \int_{-1}^1 (e^{2y} + 2e^y)dy$.
16	**(a)**	Lower limit: $-f(1 - 3x) \cdot (-3) = \frac{3(1-3x)^3}{1+(1-3x)^2}$.
17	**(d)**	$\Delta x = \frac{2}{n}$, $x_i = 1 + \frac{2i}{n}$: $\sum \sqrt{4 + (1 + \frac{2i}{n})^2}\, \frac{2}{n}$.
18	**(a)**	$L = \frac{7}{4}$, $U = \frac{15}{4}$, average $= \frac{11}{4}$.